

DAIWIK PALAVARAPU

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<https://github.com/Daiwik2005> | Portfolio:

<https://mango-match-ec1.notion.site/Daiwik-Palavarapu-Developer-Portfolio-29efecaee74580008bddd16d11ec32b7>

Professional Summary

Highly motivated and results-oriented Software Engineering student with a strong foundation in AI/ML, FinTech, and full-stack development, seeking to leverage expertise in Python, Java, and TypeScript to engineer large-scale, innovative, and resilient technology solutions. Eager to contribute to a global financial institution like JPMorgan Chase, drive digital transformation, and thrive in a collaborative, fast-paced environment.

Skills

Programming Languages: Java, Python, JavaScript, C++, TypeScript

Frameworks & Libraries: React, Next.js, HTML, CSS, Flask, Node.js, Express.js, FastAPI, Vite, Streamlit, Tailwind CSS, ShadCN UI, Framer Motion, PyTorch, Hugging Face Transformers

Databases: MySQL, MongoDB, Redis, Prisma (ORM), Mongoose (ORM), Convex (Serverless DB)

Tools & Platforms: Docker, Linux, Data Structures & Algorithms, Competitive Programming, Machine Learning, Deep Learning, LangChain, OCR (Tesseract), NumPy, JWT Authentication, Google Gemini API, Ollama (Local LLMs), OpenCV, Text-to-Speech (TTS), Google APIs, Bcrypt, System Design, Network Programming, Configuration Management, Socket Programming, Agile Methodologies, CI/CD Concepts, API Integration, Behavioral AI

Cloud & DevOps: AWS, Serverless Architectures, Cloud Technologies

Embedded / IoT: Internet of Things (IoT), Embedded Systems

Projects

Reflectify.ai: AI-Driven Behavioral Finance Platform

Conceptualized and developed an innovative AI-driven platform (Reflectify.ai) integrating behavioral psychology with financial decision-making, designed to foster mindful spending habits and improve purchase choices through intelligent reflection tools.

Technologies: React 19, TypeScript, Vite, Tailwind CSS, ShadCN UI, Framer Motion, Convex (Serverless, Database, Auth), Gemini API, Behavioral AI, Machine Learning

GitHub: <https://github.com/Daiwik2005/mindful-spend-ai>

- Led development of Reflectify.ai, an **AI-driven behavioral finance platform** utilizing **Gemini API** and **machine learning** to offer decision insights and promote mindful spending.
- Engineered a modern, responsive frontend with **React 19, TypeScript, Vite, Tailwind CSS, ShadCN UI, and Framer Motion**, ensuring a seamless and engaging user experience.

- Designed and implemented a scalable backend using **Convex** (serverless functions, database, authentication) to manage user patterns, quiz insights, and reflection logs securely.
- Integrated **behavioral psychology principles** through features like a Decision Quiz, Reflection Timeline, and Emotion-Aware Feedback, significantly influencing user purchasing decisions.
- Developed an **AI Insight Engine** to learn from user patterns, providing personalized, empathetic nudges and notifications to prevent impulsive buys and support financial wellness.

SmartPocket: AI-Powered Voice Financial Assistant

Engineered an AI-powered, voice-activated financial assistant tailored for teenagers, integrating modern web technologies with secure backend services to enhance financial literacy and budget management.

Technologies: React, Vite, Node.js, Express.js, MongoDB, Mongoose, JWT Authentication, Bcrypt, Gemini API, TypeScript, Recharts

GitHub: <https://github.com/Daiwik2005/SmartPocket>

- Developed an **AI-powered financial assistant** using **React (Vite)**, **Node.js (Express)**, and **MongoDB** to manage over 1,000 expense records with real-time tracking.
- Integrated **Gemini API** and **voice-based interaction** to improve expense logging efficiency by 40%, enhancing user experience through natural language processing.
- Implemented **JWT Authentication** and **bcrypt** for robust user security and data privacy, ensuring secure financial record management.
- Designed a scalable full-stack solution with **Mongoose ORM** and **Recharts** for intuitive data visualization, supporting efficient budget management and spending trend analysis.
- Applied principles of **gamification and behavioral psychology** to foster smart money habits and engagement for young users, aligning with FinTech innovation.

AI-Powered Document Verification System

Developed a high-accuracy OCR-based system for automated identity document validation, leveraging AI to streamline processing and significantly reduce manual effort in critical verification workflows.

Technologies: React, Node.js, Express.js, MongoDB, OCR (Tesseract), API Integration, Python

GitHub: <https://github.com/Daiwik2005/document-verification>

- Engineered an **OCR-based document verification system** achieving 99.27% accuracy, automating identity validation to enhance operational efficiency.
- Reduced manual verification effort by over 60% through **streamlined document processing and data extraction workflows** utilizing **Tesseract OCR**.
- Designed and implemented secure **MongoDB** database storage for verification records and extracted metadata, enabling efficient retrieval, audit tracking, and compliance.
- Developed a full-stack application using **React and Node.js (Express)**, with robust **API integration** for instant validation against external databases, addressing a critical need in public sector problem statements.
- Focused on **security and data integrity** in handling sensitive identity documents, building a solution capable of processing both digital and physical documents.

Win: High-Performance Web Server from Scratch

Built a lightweight, high-performance web server from first principles in Python, inspired by NGINX architecture, demonstrating core web server functionality and advanced network programming concepts.

Technologies: Python, HTTP Protocol, Network Programming, System Design, Configuration Management, Socket Programming

GitHub: <https://github.com/Daiwik2005/Win>

- Architected and implemented a **high-performance web server from scratch in Python**, adopting **NGINX-style architectural principles** to handle HTTP/1.1 requests and responses.
- Developed robust **HTTP protocol parsing** and **configuration management modules** for efficient request handling and server setup, showcasing deep understanding of network fundamentals.
- Engineered **intelligent URI routing** with longest prefix matching, enabling flexible and scalable request routing for diverse web applications.
- Implemented a **modular architecture** with custom exception handling, ensuring resilience and maintainability for core web server functionalities.
- Demonstrated strong command of **System Design, Network Programming, and low-level socket operations** crucial for building large-scale, resilient technology solutions.

Education

Bachelor of Technology – CSE- Internet of things (IOT) with proficiency in Robotics

VNR Vignana Jyothi Institute of Engineering and Technology, Hyderabad

Graduation Year: 2023-2027

BS in Data Science (Online Degree)

Indian Institute of Technology Madras

Graduation Year: Present

Intermediate

Tapasya Junior College

Graduation Year: 2021-2023